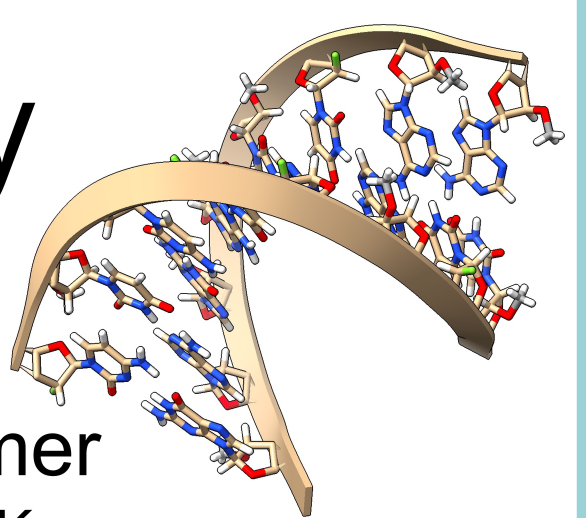
A Molecular Dynamics Approach to Study Seed-Region Modifications in siRNA



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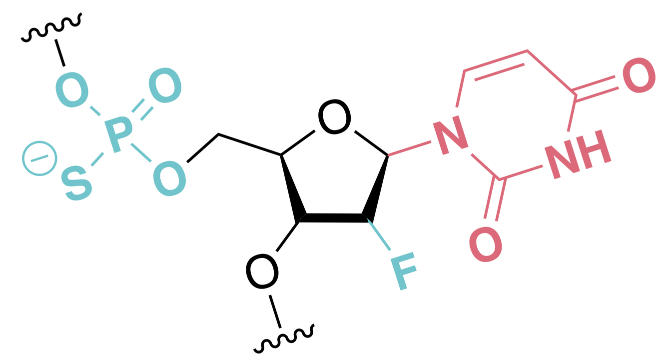
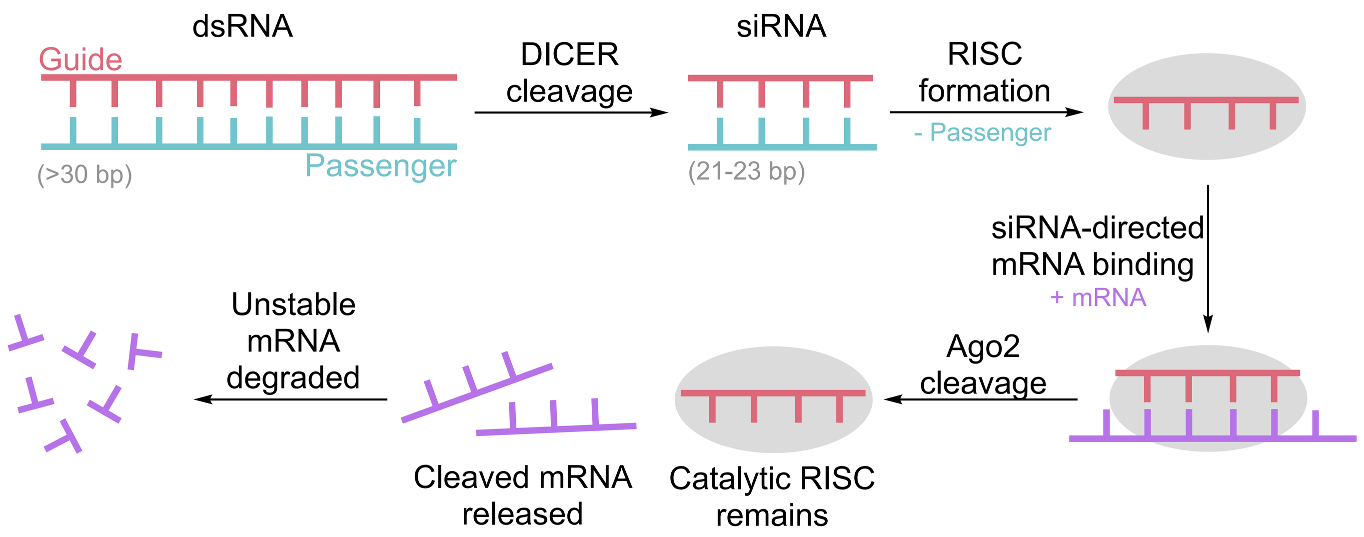
Background

RNA therapeutics offer an alternative mechanism of action to traditional small molecule drugs. Their target is dependent on **base sequence,** and their properties can be independently optimised by introducing **chemical modifications** to the backbone, sugar, and base.[1]



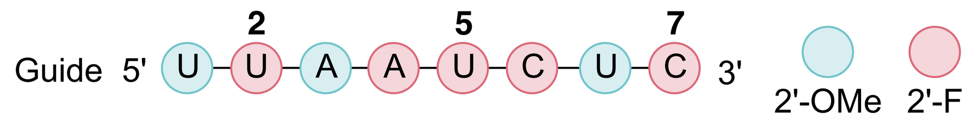
Small interfering RNAs (siRNAs) are loaded into the **RNA-induced silencing complex** to facilitate the binding of an **mRNA target** and inhibit gene expression. This is a catalytic mechanism (RNA interference) where target binding initiates at the guide strand's seed-region: residues 2-8 from the 5'-end.[2,3]



Six of seven marketed siRNA therapeutics employ chemical modifications (cm) to improve binding affinities, physiological stability, and reduce off-targets.

Molecular dynamics study of cm-siRNAs is hindered by the lack of unnatural residue force-field parameters. In response, toolkits like modXNA have been developed to calculate and assemble those missing values.[5]

This study assesses the impact of position-specific cm's on the structural dynamics, and binding free-energies, of an 8-residue siRNA sequence targeting HTT mRNA.



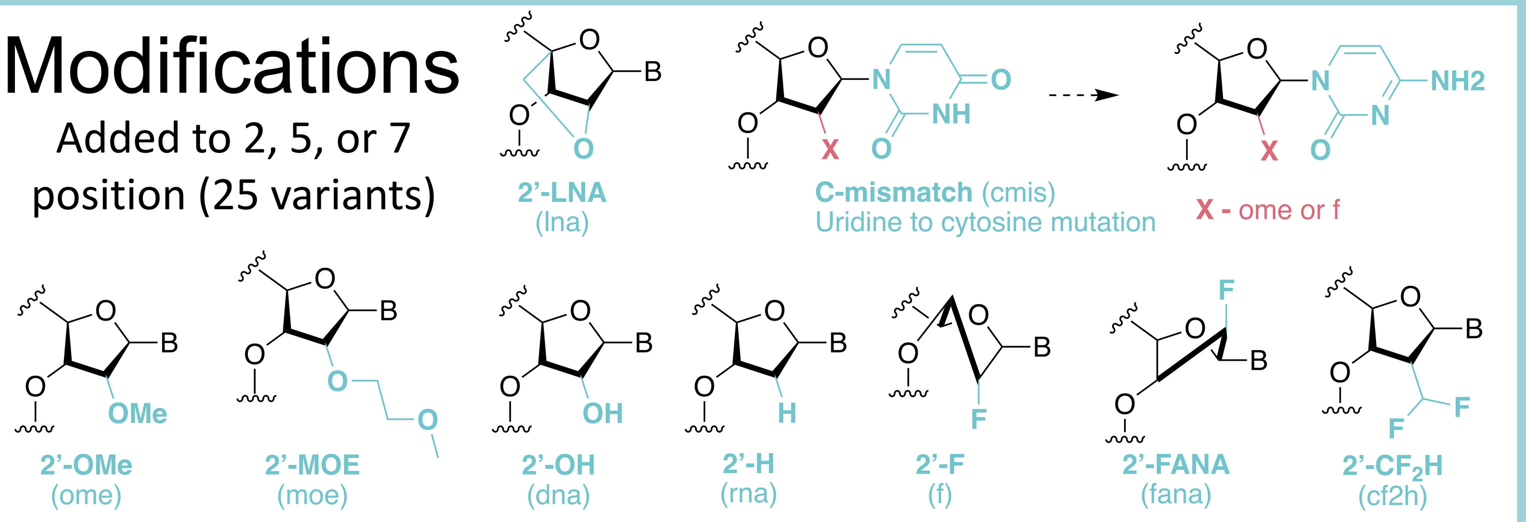
Methods

Structure analysis: RMSD, per-residue RMSF, radius of gyration, hydrogen-bond distance, pseudo- and χ-angles, intra- and inter- base-pair steps

Binding energy analysis: van der Waals, electrostatics, enthalpies, entropies, free-energies

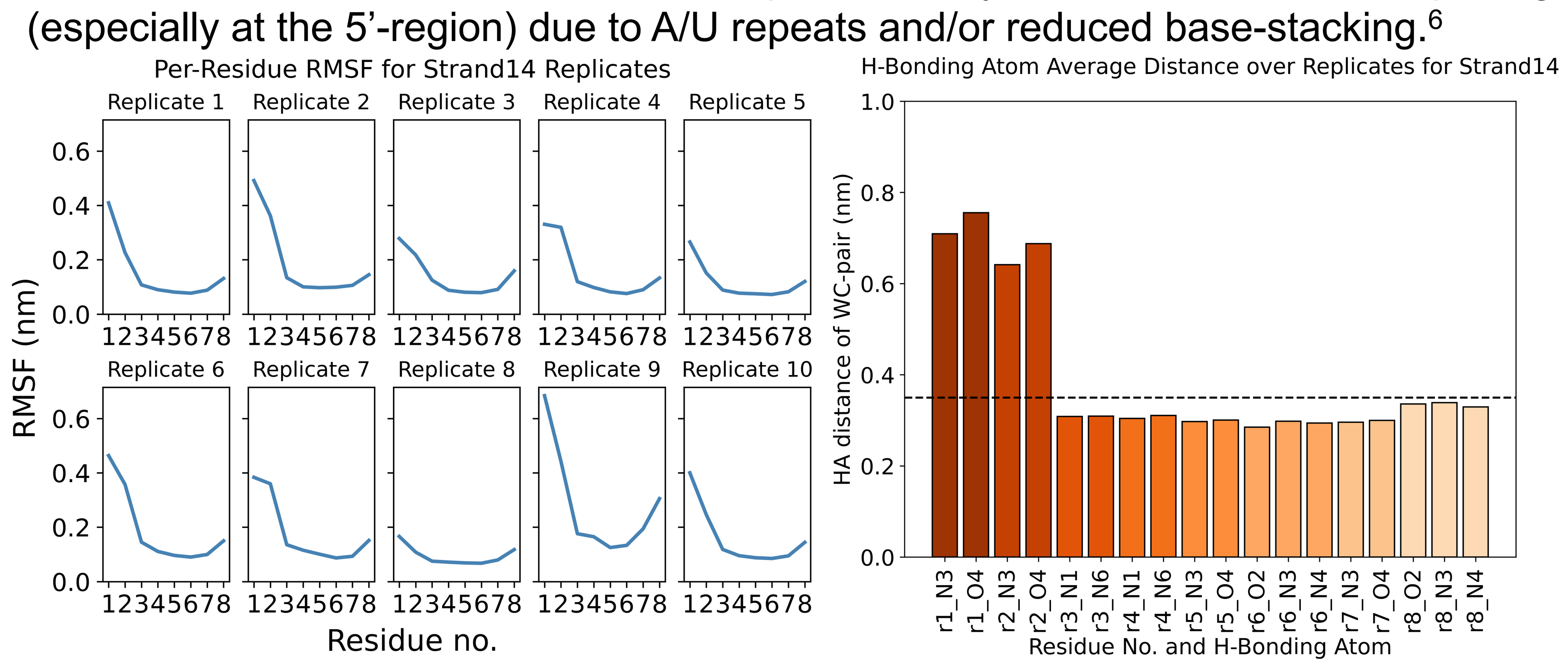
Modifications

Added to 2, 5, or 7 position (25 variants)

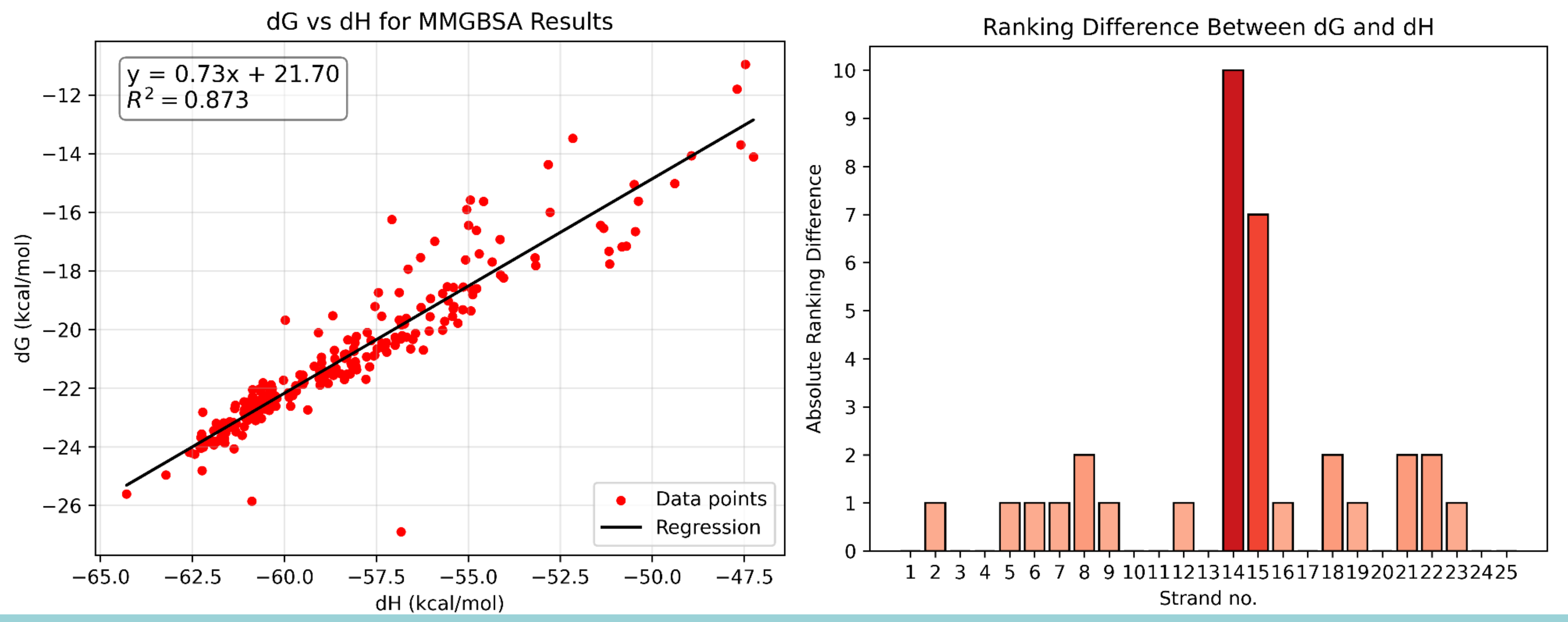


Global Analysis

Across the 25 sequence variants, RMSD values were consistent and indicative of the maintenance of duplex structure. **Per-residue RMSF's** and **hydrogen-bond distances** between Watson-Crick base-pairs convey weaker terminal base-pairing (especially at the 5'-region) due to A/U repeats and/or reduced base-stacking.[6]

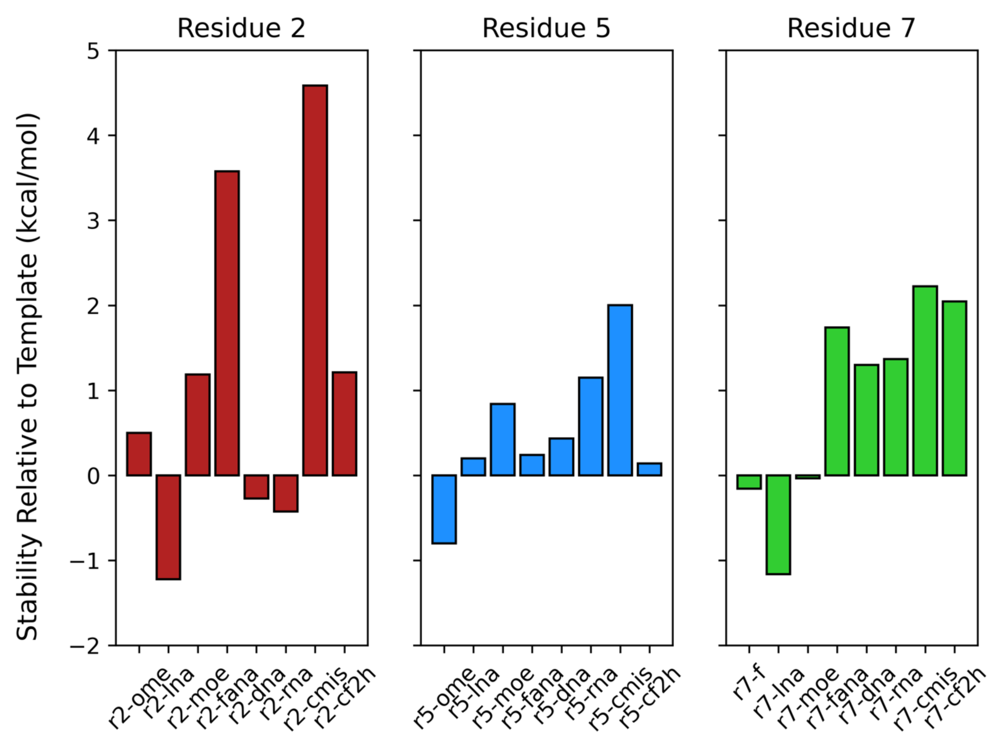


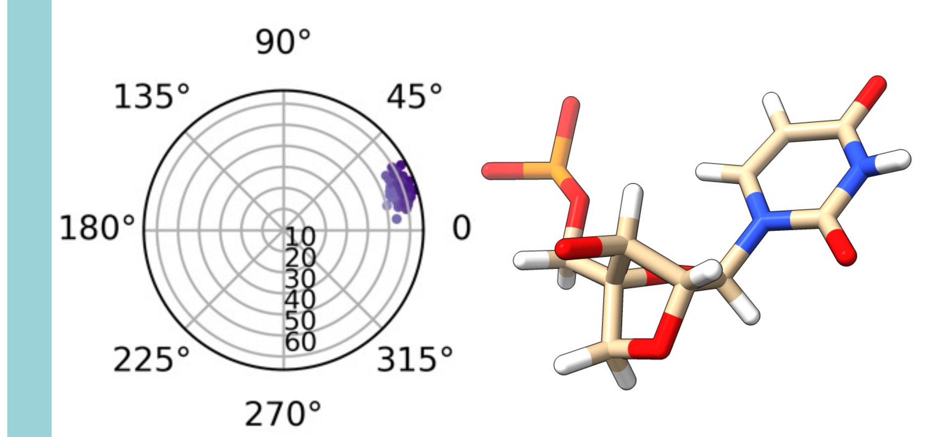
Strong agreement shown between rankings of free-energies and enthalpies from MMGBSA binding energies, however, sequences modified with **DNA** at the **2nd** (s14) and **5th** (s15) residues had a significant difference in **ranking placement.**



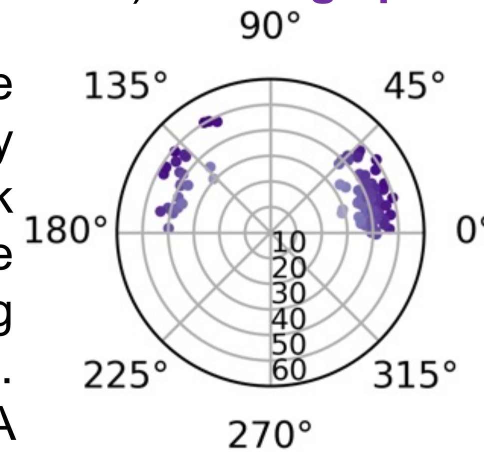
Modification Analysis

Binding affinities followed the expected literature trend for the key stabilising and destabilising residues. The unforeseen outlier was the weak binding of **MOE** variants. As the implicit solvent of MMGBSA won't capture the known crystal water interactions, lower solvation energies reduced affinity.[7]

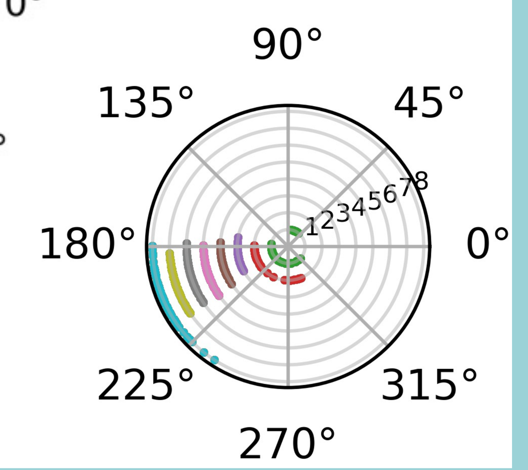




LNA's demonstrated strong binding affinities, particularly at positions **2** and **7**, due to favourable enthalpies from hydrogen-bonding and base-stacking interactions. The simulations also replicated the tight inter-base **C3'-endo pseudoangle** (favourable for A-RNA, circ.) and **large pucker amplitude** (rad.).[8]



Cytosine-mismatches were among the weakest binders: unsurprising, as they employ non-canonical Watson-Crick base-pairs. Position **2 cmis** had the most penalising binding enthalpy, along with: global instability (largest avg. RMSD), occupation of the A-RNA competing **C2'-endo pseudoangle (144° to 180°)**, and movement of the **residue 2 χ-angle** towards the syn-conformation (90° to -90°).[9]



Conclusions

- modXNA parameters were easy to generate and output stable sims. of cm-siRNAs with the expected trend in per-residue deviations and H-bond distances.
- The strongest and weakest binders at choice seed positions were predicted using this workflow, and the properties of global/local deviations, base-pair and sugar-angles, and thermodynamics, were consistent with literature observations.
- Due to the misalignment between free-energy and enthalpic rankings for DNA-modified residues, NMA-entropies were considered necessary for comparisons.
- A hybrid implicit/explicit solvent method may be required to reproduce specific water-interactions seen in the crystal-structures of modified oligonucleotides.

Future work

- An experimental component: to determine binding thermodynamics (melting points) and kinetics (on/off rates) for seed-region modifications, and compare to the aforementioned computational output.
- The incorporation of the hAgo2 protein (from the RISC) in molecular dynamics simulations, in addition to a range of cm's, to probe their effects on structure and energetics in a biologically-relevant context.
- Investigating machine-learning algorithms, and fingerprinting methods appropriate for the representation of cm-nucleotides, to predict binding affinities and thus accelerate the experimental pipeline.

Acknowledgements

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References

1. *Nat. Biotechnol.* **2017**, *35*, 238; 2. *Front. Pharmacol.* **2019**, *10*, 444; 3. *Health Sci. Rev.* **2024**, *11*, 100157; 4. *Signal Transduct. Target. Ther.* **2020**, *5*, 101; 5. *J. Chem. Theory Comput.* **2024**, *20*, 9354; 6. *J. Am. Chem. Soc.*, **2006**, 128, 6885 ; 7. *Nucleic Acids Res.*, **1998**, 26, 3694–3699; 8. Datta *et al*, *Chem. Commun.*, **2024**, 60, 13024; 9. *Nucleic Acids Res.*, **2020**, 48, e74

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